

NAND Gate

Truth table

NAND Gate		
a	b	y
0	0	1
0	1	1
1	0	1
1	1	0

Behavioral source code (nand_gate.v)

```
module nand_gate(y,a,b);  
output y;  
input a,b;  
assign y=~(a&b);  
endmodule
```

Testbench source code (nand_gate_tb.v)

```
`timescale 1ns/1ps  
  
module nand_gate_tb;  
    // 1. Creating registers for inputs and a wire for the output  
    reg test_a;  
    reg test_b;  
    wire test_y;  
  
    // 2. Instantiating the actual NAND gate module (connect pins)  
    nand_gate uut (  
        .y(test_y),  
        .a(test_a),  
        .b(test_b)  
    );  
endmodule
```

```
// 3. Generating the input signals over time
initial begin
    // Telling iverilog to dump waveform data for GTKWave
    $dumpfile("simulation.vcd");
    $dumpvars(0, nand_gate_tb);

    // Test Case 1: Both 0;
    test_a = 0; test_b = 0; #4;

    // Test Case 2: 0 and 1;
    test_a = 0; test_b = 1; #8;

    // Test Case 3: 1 and 0;
    test_a = 1; test_b = 0; #16;

    // Test Case 4: 1 and 1;
    test_a = 1; test_b = 1; #32;

    // Test Case 0: Both 0 (again);
    test_a = 0; test_b = 0; #4;

    $finish; // End simulation
end
endmodule
```

```
call C:\oss-cad-suite\environment.bat
```

fgfg

```
iverilog -o simulation.vvp nand_gate.v nand_gate_tb.v
```

fgfg

```
vvp simulation.vvp
```

Generate & check output waveform - Fig. 4(a)

```
gtkwave simulation.vcd
```

```
yosys
```

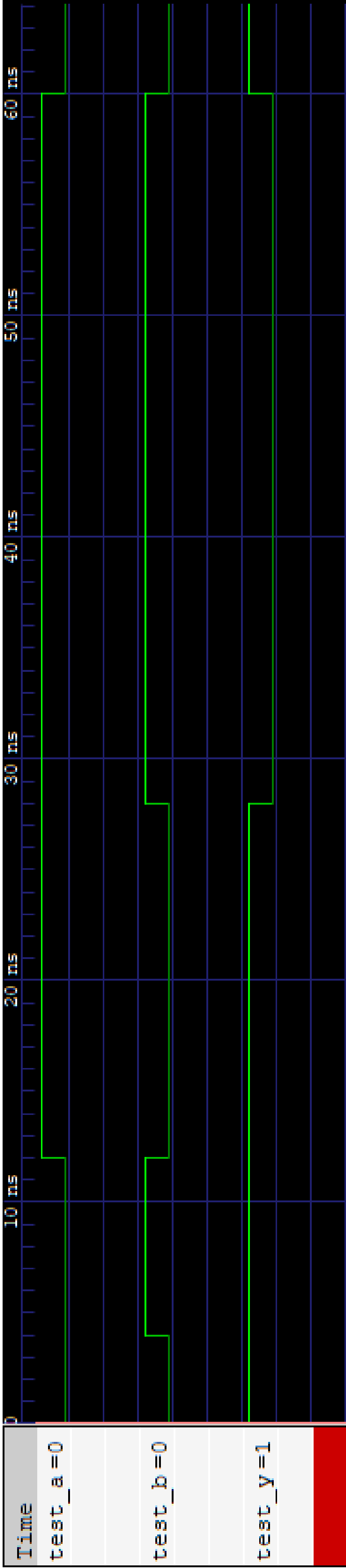


Fig. 5(a) - NAND gate output waveform

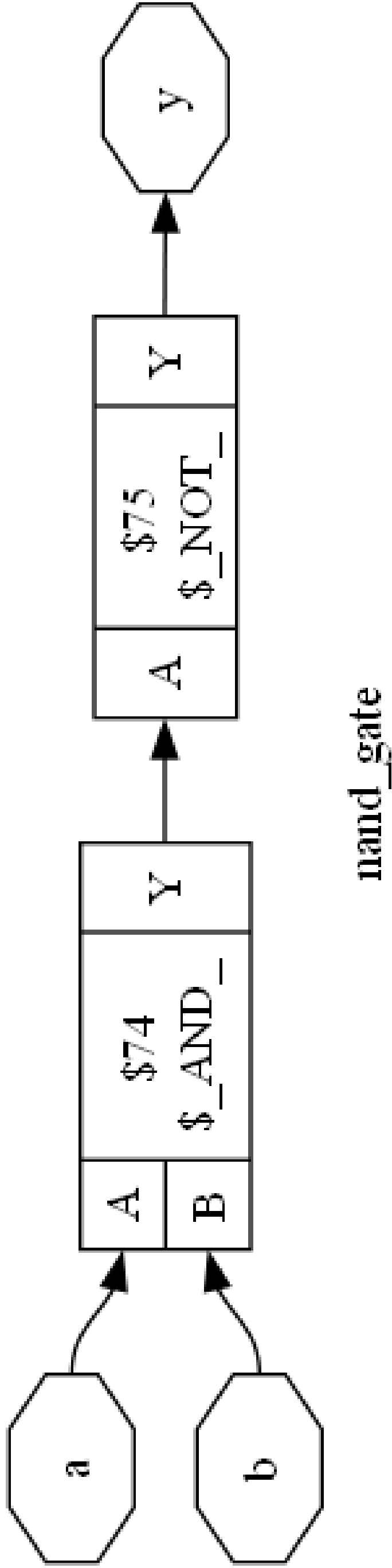


Fig. 5(b) - NAND gate generated schematic

```
/-----\
| yosys -- Yosys Open SYnthesis Suite                      |
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\-----/
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
```

```
yosys> read_verilog nand_gate.v
```

```
yosys> hierarchy -top nand_gate
```

```
yosys> proc opt techmap opt
```

```
yosys> write_verilog nand_gate_netlist.v
```

```
yosys> show
```

```
8. Generating Graphviz representation of design.
Writing dot description to 'show.dot'.
Dumping module nand_gate to page 1.
```

```
yosys> exit
```

```
[OSS CAD Suite] D:\> "C:\Program Files\Graphviz\bin\dot.exe" -Tpng show.dot -o schematic.png
```

Generate & check and_gate schematic - Fig. 5(b)

